

**Question Paper for 1st Periodical Examination (March 2026) (Session Jan-June 2026)**

Time: One and a half hours

Max Marks:30

Note: Attempt all questions.

Date: 09.03.2026(2-3:30PM)

CO	COURSE OUTCOMES
CO1	To teach the concept of language translation and the phases of compiler design.
CO2	To inculcate the knowledge of the common forms of parsers, the LL parser and the LR parser.
CO3	To demonstrate intermediate code using the technique of syntax-directed translation.
CO4	To illustrate the various optimization techniques for designing optimizing compilers.

Q No.	Questions	Marks	Target CO	Target KL
1	Answer the following questions:	1*5=5	1,2,3	
1.a	Bottom-up parsing follows which derivation?	1	CO2	L2
1.b	What is a self-hosting compiler?	1	CO1	L1
1.c	One or more objects generated by a compiler are combined into a single executable program by.....	1	CO1	L2
1.d	The first compiler was written by.....	1	CO1	L1
1.e	What is an item in LR parsing?	1	CO2	L2
2.	Implement the phases of the compiler on the following instructions: float r, h, volume; volume = (1.0/3.0) * 3.14 * r * r * h;	5	CO2	L3
3.	Write a recursive descent parser for the grammar: $S \rightarrow E$ $E \rightarrow T+E \mid T$ $T \rightarrow \text{num} * T \mid \text{num}$	5	CO2	L3
4	a. What is left recursion? How do we remove it?	2	CO2	L3
	b. Write the rule to find the First and Follow of a grammar. Show with an example.	3		
5.	Attempt any one:			
5a.	For the grammar: $S' \rightarrow S$ $S \rightarrow V;S \mid \epsilon$ $V \rightarrow \text{int id}$ Tell whether The Grammar is LL(1)? Is grammar SLR(1)?	5+5	CO2	L3
5b.	What is an operator grammar? Explain operator precedence parsing for the grammar given in question 3. Also, parse the string of your choice for the grammar.	2+8		